

ANH LE

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PROJECTS

Foundation Model Adaptation for *Thin-Structure* Segmentation

Hackathon challenge, team of 3

Fargo, ND
Jul. 2026 – Present

- Improved thin-filament segmentation from 0.60 to 0.63 Dice over full fine-tuning, by **LoRA** tuning a **SAM 2** encoder on 960 labeled solar images with under **1%** of encoder weights updated.
- Continued **self-supervised pretraining** of the SAM 2 encoder on **100k** unlabeled solar images before fine-tuning, evaluating the gain on the same held-out folds.
- Compared six models under a fixed 5-fold split, changing one factor per run to separate pretraining gains from decoder design.

Adapting a DINOv3 Foundation Model to Seismic Fault Segmentation

Self-directed research study

Fargo, ND
Aug. 2026 – Present

- Built a fine-tuning pipeline for **DINOv3** ViT-B/16 that supports frozen, **LoRA**, LayerNorm-only and full fine-tuning (**peft**), and evaluated all of them on synthetic (FaultSeg3D) and real (CRACKS) seismic data.
- Found that more trainable weights always gave higher accuracy on synthetic data: IoU 0.613 (frozen) $\rightarrow$ 0.656 (LoRA $r=4$) $\rightarrow$ 0.707 (full fine-tuning).
- Found the opposite on real field data: the lightly tuned model (LayerNorm-only, IoU 0.125) beat the fully fine-tuned one (0.109) and was $5\times$ more stable across seeds.

Zero-Shot Video Object Segmentation and Tracking on UAV Footage

Manuscript in progress

Fargo, ND
Oct. 2024 – Present

- Matched a ground-truth field sensor within **1%** using **SAM 2.1 Hiera-Large** with no training data, by prompting it for masks of the fixed ground.
- Removed drone drift by matching fixed points inside those masks across frames, dropping bad matches, and solving one correction per frame.
- Ran a second pass of the same model as the tracker, propagating one mask per floating object so centroid displacement became a velocity measurement.

YourEyes: Vision-Based Indoor Navigation for Blind Users

iOS application, team of 3

Fargo, ND
Jun. 2026 – Present

- Embedded indoor photos with frozen DINOv2 and benchmarked retrieval accuracy against open-source baselines.
- Built a backend service performing vector similarity search over the embedding database.
- Built photo-burst capture carrying heading and altitude metadata, with spoken turn-by-turn guidance.
- Ran on-device LiDAR depth sensing to raise obstacle alerts inside the same real-time capture loop.

RESEARCH EXPERIENCE

Graduate Research Assistant — ICE Lab & AI SUSTEIN, NDSU

Bathymetry Inversion from UAV Video

Fargo, ND
Oct. 2024 – Present

- Reconstructed bed elevation with R^2 of **0.85** against the surveyed bed profile, using only surface velocity derived from the frozen model's masks (**SAM 2.1 Hiera-Large**).
- Built the full path from raw video to physical result, so a general-purpose vision model, not a site-specific one, replaced a boat-based survey.
- Fed that velocity into a physics-constrained network that enforces the shallow water equations, with measured flow rate as a soft constraint.

EDUCATION

Ph.D. Civil & Environmental Engineering

North Dakota State University (NDSU), Fargo ND

Expected Dec 2027

M.S. & B.S. Civil Engineering, HCMUT, Vietnam

2020 / 2018

TECHNICAL SKILLS

- **Languages:** Python, MATLAB, SQL, Bash
- **Deep learning:** PyTorch, JAX, Hugging Face Transformers
- **Imaging & CV:** OpenCV, scikit-image, PIVlab, SAM, DINO
- **Data & infrastructure:** PostgreSQL + pgvector, Docker, Git, SLURM